An *ab-initio* Diagrammatic Monte Carlo prototype for the LiF electron polaron

Samuel Recker

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Abstract

This note fixes the **goals**, deliberate **non-goals**, and **deferred** extensions of the `rmc-ab-initio` project: a diagrammatic Monte Carlo (DiagMC) prototype, built on the `ironmc` engine, that replaces the analytic Fröhlich electron–phonon vertex $|g(\mathbf{q})|^2 \propto \alpha/q^2$ with the *real* electron–phonon matrix elements $g(\mathbf{k}, \mathbf{q}, \nu)$ obtained from density-functional perturbation theory and Wannier interpolation (Quantum ESPRESSO + EPW). The first validation material is the electron polaron in LiF, a wide-gap ionic insulator that sits close to the textbook Fröhlich limit and therefore provides a built-in analytic cross-check. The scope below is the *first* prototype; Section 4 records what is intentionally postponed.

1 Overview

`ironmc` is a Rust workspace of DiagMC crates that sample the imaginary-time Green's function of a polaron. This project adds `rmc-diagmc`, a state-generic DiagMC engine exposing a `Model` trait (electron dispersion, phonon dispersion, complex e–ph vertex, phonon-momentum proposal), plus a standalone *ab-initio* consumer `rmc-ab-initio`. Two models implement the trait: the closed-form *analytic* Fröhlich model, and the *ab-initio* model whose material-specific electron–phonon (e–ph) vertex $g(\mathbf{k}, \mathbf{q}, \nu)$ is read from a real DFT/DFPT + Wannier calculation. The ground-state energy is read off the large-τ decay of $G(\tau)$; there is no self-consistent iteration.

The data pipeline is

$$\underbrace{\text{pw.x, ph.x}}_{\text{Quantum ESPRESSO}} \longrightarrow \underbrace{\text{epw.x}}_{\text{Wannier e–ph tensors}} \longrightarrow \underbrace{\text{Rust reader + reconstruction}}_{\texttt{crates/apps/rmc-ab-initio}} \longrightarrow \text{DiagMC.}$$

DFPT supplies the phonons, the Born effective charges $\mathbf{Z}^*_\kappa$, and the electronic dielectric tensor $\boldsymbol{\varepsilon}_\infty$; EPW supplies the e–ph coupling in the localized Wannier representation $g(\mathbf{R}_e, \mathbf{R}_p)$, from which $g(\mathbf{k}, \mathbf{q}, \nu)$ is recovered by an inverse Fourier transform.

Why LiF. Rocksalt LiF has a single, non-degenerate conduction-band minimum at Γ, one dominant longitudinal-optical (LO) branch, and an intermediate-to-strong Fröhlich coupling $\alpha \approx 4.8$. It is therefore the cleanest first material: nearly the textbook Fröhlich case, with a genuine analytic limit to validate against.

Units. The ab-initio model works in *physical* units — energies in eV, imaginary time in eV^{-1}. The electron dispersion $\varepsilon(\mathbf{k})$ is the real Wannier-interpolated conduction band from $H(\mathbf{R})$ (not a parabola), and the phonons are the real branches $\omega_{\mathbf{q}\nu}$ from the polar-restored dynamical matrix,

with *all* modes retained. The Fröhlich coupling is still computed from the macroscopic constants as a cross-check,

$$\alpha \;=\; \frac{e^2}{\hbar}\sqrt{\frac{m^*}{2\hbar\omega_{\mathrm{LO}}}}\left(\frac{1}{\varepsilon_\infty}-\frac{1}{\varepsilon_0}\right) \;\approx\; 4.76 \quad \text{(LiF)}, \tag{1}$$

which fixes the weak-coupling target $E^{(2)} \approx -\alpha\,\hbar\omega_{\mathrm{LO}}$ against which the vertex scale is checked (Section 8, P8).

2 Goals

G1 — Real vertex. Drive the DiagMC sampler with the *ab-initio* e–ph vertex in place of the analytic one. The vertex is decomposed as

$$g_{mn\nu}(\mathbf{k},\mathbf{q}) \;=\; g^{\mathcal{S}}_{mn\nu}(\mathbf{k},\mathbf{q}) \;+\; g^{\mathcal{L}}_{mn\nu}(\mathbf{k},\mathbf{q}), \tag{2}$$

the short-range (Wannier-interpolated) part plus the long-range polar part.

G2 — Generic engine, native model. Factor the sampler into a state-generic `rmc-diagmc` engine (`Runner`, `Update`, `MetropolisKernel`, decay estimator) behind a `Model` trait. The ab-initio model is a *native* implementation of that trait — real $\varepsilon(\mathbf{k})$, $\omega_{\mathbf{q}\nu}$, complex $g(\mathbf{k},\mathbf{q},\nu)$, and a $|g|^2$-weighted phonon-momentum proposal — *not* a reweighting of the Fröhlich envelope. The Fröhlich model is a second implementation of the same trait, which validates the engine at rung 0 (Section 5).

G3 — Correct polar long range. Restore the long-range Fröhlich $1/q$ divergence that dominates the LiF polaron. EPW was run with `lpolar=.true.`, which *subtracts* the long-range part before Wannierization — from *both* the e-ph vertex and the dynamical matrix — so $g^{\mathcal{S}}$ from the tensor is short-ranged and both the vertex term $g^{\mathcal{L}}$ (Eq. (3)) and the dynamical-matrix term $D^{\mathcal{L}}$ (Eq. (4)) must be added back analytically. The latter is what restores the physical TO frequency and the LO–TO splitting; without it the phonon zero-point factor $(2M_\kappa\omega)^{-1/2}$ in $g^{\mathcal{L}}$ is mis-scaled (Section 8, P7). *Done:* both implemented as from-scratch ports of EPW's `rgd_blk_epw` and `rgd_blk` (Ewald parameters $\Lambda_{\mathrm{Ew}} = 1$, $g_{\max} = 14$, $e^2 = 2$ Ry) and validated to machine precision against the coarse Bloch `.epb` and dynamical matrix `dynq` (Section 8, P6–P7).

G4 — Clean Rust reader. Implement the EPW-file readers and the Wannier reconstruction from scratch in Rust, with no Python at runtime. The existing Python post-processor (`patt126-ab-initio-diagmc`) is used as a *reference only*, not copied; the rewrite also fixes a latent index assumption in the reference.

G5 — Two-front validation. Benchmark the LiF electron polaron against *both* (i) Fröhlich-limit DiagMC energetics at α_{LiF} (Mishchenko *et al.*, 2000), and (ii) *ab-initio* LiF polaron numbers (Sio, Verdi, Poncé, Giustino, 2019).

The long-range term used for G3 (Verdi–Giustino form) is

$$g^{\mathcal{L}}_{mn\nu}(\mathbf{k},\mathbf{q}) \;=\; i\,\frac{4\pi e^2}{\Omega}\sum_{\kappa}\left(\frac{\hbar}{2M_\kappa\omega_{\nu\mathbf{q}}}\right)^{1/2}\sum_{\mathbf{G}\neq-\mathbf{q}}\frac{(\mathbf{q}+\mathbf{G})\cdot\mathbf{Z}^*_\kappa\cdot\mathbf{e}_{\kappa\nu}(\mathbf{q})}{(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\varepsilon}_\infty\cdot(\mathbf{q}+\mathbf{G})}\,e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_\kappa}\,e^{-\frac{(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\varepsilon}_\infty\cdot(\mathbf{q}+\mathbf{G})}{4\Lambda}}\left[U_{\mathbf{k}+\mathbf{q}}U^\dagger_{\mathbf{k}}\right]_{mn}, \tag{3}$$

with Ω the cell volume, $M_\kappa, \boldsymbol{\tau}_\kappa$ the atomic masses and positions, $\mathbf{e}_{\kappa\nu}$ the phonon eigenvectors, and Λ an Ewald range-separation parameter. The final factor $[U_{\mathbf{k}+\mathbf{q}} U_{\mathbf{k}}^\dagger]_{mn}$ is the electron-band rotation into the smooth (Wannier) gauge; at the Γ-slice ($\mathbf{k} = \Gamma$, single conduction band) it reduces to the conduction-band Bloch overlap $\langle u_{\Gamma+\mathbf{q}} | u_\Gamma \rangle$. Restoring it is essential to the exact short/long decomposition of Eq. (2); see Section 8, P6. As $q \to 0$ the term behaves as $1/q$, reproducing the Fröhlich tail $|g_F(\mathbf{q})|^2 \propto \alpha/q^2$.

The *same* dipole physics enters the dynamical matrix: EPW likewise removes the rigid-ion (dipole–dipole) long-range from D before Wannierization, so the interpolated phonons are short-ranged — the optical triplet stays degenerate at Γ at an unphysical frequency with no LO–TO splitting. The term to add back (Gonze form, EPW `rgd_blk`) is, with $\mathbf{Q} \equiv \mathbf{q} + \mathbf{G}$,

$$D^{\mathcal{L}}_{\kappa\alpha,\kappa'\beta}(\mathbf{q}) \;=\; \frac{4\pi e^2}{\Omega} \sum_{\mathbf{G}} \frac{e^{-\mathbf{Q}\cdot\boldsymbol{\varepsilon}_\infty\cdot\mathbf{Q}/4\Lambda}}{\mathbf{Q}\cdot\boldsymbol{\varepsilon}_\infty\cdot\mathbf{Q}} \,(\mathbf{Q}\cdot\mathbf{Z}^*_\kappa)_\alpha \,(\mathbf{Q}\cdot\mathbf{Z}^*_{\kappa'})_\beta \, e^{i\mathbf{Q}\cdot(\boldsymbol{\tau}_\kappa - \boldsymbol{\tau}_{\kappa'})} \;-\; \delta_{\kappa\kappa'}\, S_{\kappa\alpha\beta}, \tag{4}$$

where the self-term $S_{\kappa\alpha\beta} = \sum_{\kappa''} D^{\mathcal{L}}_{\kappa\alpha,\kappa''\beta}(\mathbf{0})$ (the same sum at $\mathbf{q} = 0$, summed over the second atom) is subtracted on the diagonal blocks to enforce the acoustic sum rule. Diagonalizing $D^{\mathcal{S}} + D^{\mathcal{L}}$ (mass-reduced) gives the physical branches $\omega_{\mathbf{q}\nu}$ and eigenvectors $\mathbf{e}_{\kappa\nu}(\mathbf{q})$ that enter Eq. (3).

3 Non-goals (deliberate scope of the first prototype)

Several early non-goals of the *first* prototype have since been **lifted**: the Γ-slice vertex, the single Einstein branch, the "no tensor compression" rule, the reweighted sampler, and the Fröhlich-unit engine constants. The model now uses the full electron-momentum vertex $g(\mathbf{k}, \mathbf{q}, \nu)$, all phonon branches, an SVD-compressed Wannier tensor evaluated on a grid, a native $|g|^2$ momentum proposal, and physical units. What remains deliberately out of scope:

N1 — Electron polaron only. The single, non-degenerate conduction-band minimum at Γ. The hole/valence polaron (degenerate top valence band) is out of scope.

N7 — No GPU path. The ab-initio model targets the CPU driver only; the GPU (`#[cube]`) mirror is not extended to it.

N8 — Single material. LiF is the sole validation target; this is not yet a general multi-material framework.

(The retired non-goals keep their original labels N2–N6 where referenced elsewhere; they are now realized features, recorded as done in Section 4.)

4 Deferred to future work

Done. The full-$\mathbf{k}$ vertex with SVD compression and on-grid evaluation (former N2/N4), all dispersive branches $\omega_\nu(\mathbf{q})$ (N3), the non-parabolic Wannier dispersion $\varepsilon(\mathbf{k})$ via the generic `Model` trait (N6), and the native $|g|^2$ momentum sampler (N5) are now *implemented*.

Remaining.

D1 — Grid convergence. The vertex is tabulated on a uniform n^3 grid; the discrete BZ sum underweights the near-Γ $1/q^2$ polar region, so the polaron energy converges from above with a

leading $1/n$ error (at n=8 the formation energy is $E_0 \approx -0.122$ eV against a converged target near -0.4 eV). The ladder $n = 20, 32, 40, 50$ with a linear $1/n \to 0$ extrapolation (Section 6) is in progress; its dominant cost is the vertex-cache memory (Section 7), not the sampler.

D2 — Hole polaron. The degenerate valence-band-maximum polaron (lifting N1), needing a multi-band, degenerate (matrix-product sign-fix) treatment.

D3 — GPU path. A `#[cube]` GPU implementation of the ab-initio model (lifting N7), guarded by the existing lockstep parity tests.

D4 — More materials. Generalize beyond LiF once the LiF prototype is validated (lifting N8).

5 Validation ladder

The goals of Section 2 are checked in rungs of increasing physical content:

1. **Engine (rung 0).** The analytic Fröhlich model, run through `rmc-diagmc` and the decay estimator, reproduces Mishchenko *et al.* (2000): $E_0(\alpha{=}1) = -1.013$ and $E_0(\alpha{=}2) = -2.08$ within 3σ, sign $\equiv 1$. This validates the sampler, estimator, and sign tracking independently of the ab-initio vertex.

2. **Vertex scale.** The ab-initio second-order energy $E^{(2)}(\Gamma)$ recovers the Fröhlich weak-coupling target $-\alpha\,\hbar\omega_{\mathrm{LO}}$ (up to the coarse-grid underestimate), confirming the $(2M_\kappa\omega)^{-1/2}$ vertex normalization (Section 8, P8).

3. **Compression convergence.** The all-orders E_0 is certified against the compression *in the sampler itself*: the production rank budget must agree with the uncompressed vertex to within a stated tolerance, $\Delta_{\mathrm{rank}} = E_0(\mathrm{budget}) - E_0(\mathrm{uncompressed})$, and the measured Δ_{rank} is quoted as a systematic on every production number (Section 6).

4. **Material comparison.** The grid-ladder formation energy against FEP-DMC (Luo, Park, Bernardi, 2025): $E_0(20^3) = -0.248$ eV before extrapolation and -0.408 eV in the $1/n \to 0$ limit, with Sio *et al.* (2019) as a secondary reference — pending the ladder runs (D1).

6 Discretization, compression, and the thermodynamic limit

The object the sampler actually solves is not the continuum polaron Hamiltonian but its restriction $\hat{H}_n$ to a uniform n^3 Brillouin-zone grid shared by electrons and phonons. The grid is closed under momentum addition, so every diagram's momentum bookkeeping stays on it; the per-arc momentum measure is the lattice normalization $\frac{1}{N_q}\sum_{\mathbf{q}\neq\Gamma}$ with $N_q = n^3$ (matching the standard first-principles self-energy convention), and the DiagMC solves each $\hat{H}_n$ *numerically exactly*. All physical statements are then statements about the sequence $E_0(n)$: because the discrete sum underweights the $1/q^2$ polar region within a $1/n$-neighbourhood of Γ, $E_0(n)$ converges from above with a leading error linear in $1/n$, and the thermodynamic limit is read off a linear $1/n \to 0$ extrapolation over a ladder of grids ($n = 20 \ldots 50$ for LiF) — the same methodology as FEP-DMC, whose LiF benchmarks are $E_0(20^3) = -0.248$ eV and -0.408 eV extrapolated.

Compression as a controlled, quoted approximation. The short-range vertex is compressed to a global rank budget (the largest singular values across all band pairs); the long-range polar term is always exact. The acceptance criterion is *never* element-wise reconstruction error — it is convergence of the physical observable, certified in the sampler itself:

$$\Delta_{\text{rank}} \;=\; E_0(\text{budget}) - E_0(\text{uncompressed}),$$

measured by two otherwise-identical DMC runs and quoted as a systematic error alongside the statistical one on every production result. Two empirical lessons fixed this protocol. First, the cheap proxy — the second-order energy $E^{(2)}$ at $\mathbf{k} = \Gamma$ — systematically *overestimates* the true sampler systematic by a factor of 3–10, because the sampler averages the compression error over the whole zone while the proxy probes a single $\mathbf{k}$-row; the proxy is useful for scanning, but only the in-sampler measurement decides. Second, convergence with the global budget is *non-monotonic* (singular values rank global tensor weight, not importance for a given observable), so "two nearby budgets agree" is not evidence of convergence — only the comparison against the uncompressed vertex is. For the current LiF dataset the certified production budget is 800 of 1600 components with $\Delta_{\text{rank}} = 1.0 \pm 0.8$ meV; the FEP-DMC-sized budget (≈ 300) measured a genuine 4.6 ± 0.8 meV and was rejected.

Dataset provenance. The DFPT/Wannier inputs use $8 \times 8 \times 8$ coarse k- and q-grids, matching FEP-DMC's LiF setup. Upgrading from the initial $6 \times 6 \times 6$ deepened $E_0(8^3)$ by ~ 11 meV — a pure interpolation-quality correction, worth recording because it is the same order as the compression and statistical errors being controlled above. One physical subtlety appeared with the finer grid: it contains the W star, where the two lowest conduction bands are exactly degenerate, making per-band comparisons gauge-ambiguous there (the round-trip validation skips those points; the sampler itself is unaffected, since it consumes gauge-invariant $|g|^2$ combinations).

7 Performance: the ab-initio vertex cache

The full electron-momentum vertex — the former Γ-slice non-goal (N2), now lifted — cannot be reduced to the scalar $|g(\Gamma, \mathbf{q}, \text{LO})|^2$. Instead the Wannier e–ph tensor is compressed by a per-band-pair truncated SVD into rank-r factors $u_g(\mathbf{k})$, $v_g(\mathbf{q}, \nu)$, tabulated once on the q-grid; a vertex evaluation is then the rank-r contraction

$$g(\mathbf{k}, \mathbf{q}, \nu) \;=\; \sum_{g=1}^{r} s_g \, u_g(\mathbf{k}) \, v_g(\mathbf{q}, \nu) \;+\; g^{\mathcal{L}}(\mathbf{k}, \mathbf{q}, \nu),$$

called millions of times per DiagMC chain, so its cost sets the wall-clock. On an 8^3 grid at $r \sim 150$ the factor tables are ≈ 0.18 GiB — far larger than cache — so the memory layout of the component axis g, not the arithmetic, dominates.

Cache-hostile layout (before). Storing the factors component-major, $u[g][\mathbf{k}]$ and $v[g][\mathbf{q}][\nu]$, places successive components of a single vertex sum ~ 49 KB apart, so each of the r terms is a separate cache miss and the inner contraction is bound entirely by memory latency.

Fix: transpose the component axis inward. Storing the factors $\mathbf{k}$-major / $(\mathbf{q}, \nu)$-major — $u[\mathbf{k}][g]$, $v[\mathbf{q}][\nu][g]$ — makes the r components of one vertex evaluation contiguous, so the contraction streams two cache-line runs instead of striding r scattered lines. The precompute is reorganized to parallelize over $\mathbf{k}$ (rather than over g) so it writes this layout directly.

Effect. For the n_{sv} convergence sweep (500k steps $\times$ 10 chains, 8^3 grid), a single rank drops from 258 $\rightarrow$ 23 s at r=50 and 1033 $\rightarrow$ 54 s at r=100 — an 11–19$\times$ wall-clock reduction that grows with r. The evaluation is now memory-bandwidth bound on the factor tables; the O(order) phase recomputation in the decay estimator is a smaller secondary term, left for later.

8 Practical pitfalls and fixes

Turning the EPW run into a *usable* vertex, and then a trustworthy DiagMC number, meant working through several failures whose symptoms were misleading. P1–P7 are data-preparation (A0/A1); P8–P9 are the DiagMC consumer (Phase B). They are recorded here so they are not rediscovered; the blow-by-blow A0/A1 diagnosis, with reproducing commands, lives in `datasets/LiF/EPH-NAN-DIAGNOSIS.md`.

P1 — Ultrasoft pseudopotentials give an all-`NaN` e–ph tensor (root cause). The first A0 deck used ultrasoft pseudopotentials (SSSP-efficiency GBRV). EPW's electron–phonon matrix elements are only supported with *norm-conserving* pseudopotentials — the ultrasoft augmentation charge is not carried into the coupling. The coupling came out finite *only* at $\mathbf{q} = \Gamma$ (the trivial reference q, which needs no symmetry rotation) and `NaN` at every other $\mathbf{q}$; the Bloch$\rightarrow$Wannier sum then spread the `NaN` to 100% of `LiF.epmatwp`. Because `ph.x` *does* support ultrasoft, the phonons, dynamical matrix, $\mathbf{Z}^*_\kappa$ and $\boldsymbol{\varepsilon}_\infty$ were all correct — only the e–ph coupling was wrong, which is why the fault was easy to misattribute. *Fix:* switch Li and F to norm-conserving Pseudo Dojo ONCV (scalar-relativistic PBE). This raises the required plane-wave cutoff (`ecutwfc` : 50 $\rightarrow$ 96 Ry, `ecutrho` : 400 $\rightarrow$ 384 Ry, i.e. the norm-conserving 4$\times$), which makes `ph.x` $\sim$3–4$\times$ slower (the wavefunctions move from the small smooth FFT grid onto the dense grid) — an unavoidable cost of the correct pseudopotential family. With norm-conserving pseudos the tensor is finite at every $\mathbf{q}$, and $\boldsymbol{\varepsilon}_\infty$, $\mathbf{Z}^*_\kappa$ are essentially unchanged.

P2 — "Right size, wrong values": a silent failure. The broken run finished cleanly and wrote `LiF.epmatwp` at exactly the expected byte size; the only hint was an easily-dismissed `IEEE_INVALID` note at exit. The file was 100% uninitialised-memory `NaN` (a uniform garbage word that even *changed* between reruns). An earlier "A0 done" had been declared on file size alone, without ever checking the tensor *values*. *Fix:* the pipeline (`run_pipeline.nu`) now runs an explicit finiteness check on `LiF.epmatwp` and aborts loudly on `NaN`, so a broken run can no longer masquerade as a good one.

P3 — Two dead ends from assuming an upstream bug. Before the pseudopotential cause was found, two "EPW bug" hypotheses were pursued and *both failed*: (i) widening the Wannier outer window `dis_win_max` to enclose every band made the coarse tensor *worse*, not better; (ii) a bounds-guard patch to EPW's `gmap_sym` routine (a genuine latent out-of-bounds read) compiled in cleanly yet left the output byte-identical — so that read was not the cause. The lesson, recorded explicitly: suspect the *inputs* (here, an unsupported pseudopotential family) before patching upstream code.

P4 — The reference `.epb` reader has the wrong record layout. The Python reference (`patt126`) reads the coarse Bloch `.epb` file assuming a single band count throughout the record. The real record interleaves a *20-band* eigenvalue array with a *15-band* coupling array (of which only a 5×5 block is populated); a naïve parse silently mis-aligns and yields a *false positive* — a "the coarse data looks fine" reading that briefly sent the debugging down a wrong path. *Fix:* the true record layout was reverse-engineered and byte-verified — a *single* unformatted Fortran record (4-byte length markers) $[\, n_{qc},\ \mathbf{x}_{qc}(3, n_{qc}),\ \varepsilon(20, n_k),\ D(6, 6, n_{qc}),\ g(15, 15, n_k, 6, n_{qc}),\ \mathbf{Z}^*, \boldsymbol{\varepsilon}_\infty \,]$,

all little-endian and column-major, one file per pool with the pool-local n_k recovered from the record length. Two indexing traps: the coupling's mode axis sits *interior* (stride $n_k \cdot 6$, not $6 \cdot n_k$), and the conduction-band minimum is kept-band local index 0 in the coupling array but eigenvalue index 5 in ε. The Rust reader uses this directly and decodes only the needed conduction-band slice on demand.

P5 — epw coarse-grid prerequisites (reproducibility). Several setup constraints are load-bearing and non-obvious: the `nscf` k-points must be an *explicit* Γ-centred crystal list (an automatic, half-shifted Monkhorst–Pack mesh breaks EPW's $\mathbf{k}+\mathbf{q}$ folding); the whole `scf/ph/nscf/epw` chain must use *one* Quantum ESPRESSO version (a mixed-version `phsave` corrupts the heap); and EPW needs a large stack (`ulimit -s unlimited`). Finally, changing the pseudopotentials shifts the band edges ($\sim$0.15 eV here), so the disentanglement windows (`dis_froz_max`, `dis_win_max`) must be re-tuned against the new `nscf` eigenvalues — otherwise the frozen conduction-band minimum, which must stay exact, can slip out of its window.

P6 — The polar term lives in the Cartesian displacement basis, and carries an electron-band rotat
Two subtleties in Eq. (3) decide whether the split $g = g^{\mathcal{S}} + g^{\mathcal{L}}$ of Eq. (2) actually closes, and both were found by round-tripping against the coarse `.epb` (which stores the *full*, pre-subtraction coupling). *(i) Basis.* EPW subtracts $g^{\mathcal{L}}$ from the coupling in the *Cartesian atomic-displacement* basis ($\kappa\alpha$, i.e. per atom and Cartesian direction), *before* the phonon-eigenvector projection $\mathbf{e}_{\kappa\nu}(\mathbf{q})$ and the $(\hbar/2M_\kappa\omega)^{1/2}$ mode factor. So the reconstructed short-range part and the re-added $g^{\mathcal{L}}$ must both be formed in that basis and summed there; projecting onto branches first is inconsistent. The mode factor and eigenvector projection are applied afterwards by the consumer. *(ii) Gauge.* The band-rotation factor $[U_{\mathbf{k}+\mathbf{q}}U_{\mathbf{k}}^{\dagger}]_{mn}$ — at the Γ-slice the conduction-band Bloch overlap $\langle u_{\Gamma+\mathbf{q}}|u_\Gamma\rangle$ — is easy to drop under the "vertex depends on $\mathbf{q}$ only" reduction (both electron legs rotated with U_Γ, overlap $\equiv 1$). Dropping it puts the polar term in the wrong electron gauge: near Γ the reduced sum still overshoots $\|g\|$ by $\sim 9\%$, and the overlap itself falls from ≈ 0.94 at the first coarse shell to 0 at the zone boundary as the conduction-band character changes with $\mathbf{k}$. Restoring the exact bra ($u_{\Gamma+\mathbf{q}}$) and the overlap makes the decomposition exact. *Result:* against all 215 non-zero coarse $\mathbf{q}$, $g_{\text{full}} = g^{\mathcal{S}} + \langle u_{\Gamma+\mathbf{q}}|u_\Gamma\rangle\, g^{\mathcal{L}}$ to machine precision — gauge-invariant cosine = 1.000000 and $|\text{ratio} - 1| \sim 3 \times 10^{-7}$ (the `.epb` Kohn–Sham states carry an arbitrary per-$\mathbf{k}$ phase, so the check is on direction and magnitude, not the raw complex difference). The reduced $\mathbf{q}$-only vertex remains the object the sampler consumes; the exact bra + overlap is used only to prove the reconstruction and the polar term against real data.

P7 — The dynamical matrix carries the same polar long-range. The subtraction of P6 is not confined to the e-ph vertex: EPW removes the rigid-ion dipole–dipole term from the dynamical matrix as well (`rgd_blk`), so the Wannier-interpolated phonons are short-ranged — the optical triplet stays degenerate at Γ at an unphysical 11.95 THz with no LO–TO splitting, where the physical TO is 8.12 THz (agreeing with `ph.x`). This is easy to overlook because the phonon *eigenvectors* are barely affected (the optical modes are symmetry-fixed Li–F relative motion), but the *frequency* enters the vertex through the zero-point factor $(2M_\kappa\omega_{\mathbf{q}\nu})^{-1/2}$, so a short-range ω mis-scales $|g|^2$ by a factor of order two. *Fix:* port EPW's `rgd_blk` (Gonze form, Eq. (4)) – the $\mathbf{q} \neq 0$ dipole–dipole sum over all atom-pair blocks plus the $\mathbf{q}$=0 self-term on the diagonal blocks – and add it back before diagonalizing. *Validated* at two levels: the reconstructed long-range term equals coarse-`dynq` minus the short-range force constants *element-by-element* (ratio 1.000), and all six branches then reproduce the DFPT `dynq` at the coarse q ($[2.76, 2.76, 5.23, 8.12, 8.12, 18.45]$ THz, LO = 18.45) — the phonon

analog of the P6 round-trip. The restored $\omega_{\mathrm{LO}}(\Gamma) = 18.67$ THz gives a Lyddane–Sachs–Teller ratio $\varepsilon_0/\varepsilon_\infty = (\omega_{\mathrm{LO}}/\omega_{\mathrm{TO}})^2 \approx 5.3$, i.e. $\varepsilon_0 \approx 11$ (experiment ≈ 9).

P8 — The phonon quantization factor was dropped in the branch projection (Phase B). Distinct from P7 (where the *frequency* was wrong): here $\omega_{\mathbf{q}\nu}$ was correct, but the consumer projected the Cartesian coupling onto phonon branches with only the eigenvector $\mathbf{e}_{\kappa\nu}$ and the Ry→eV conversion, *omitting* the $(2M_\kappa\omega_{\mathbf{q}\nu})^{-1/2}$ zero-point factor of Eq. (3) entirely. Every $|g|$ was then $\sim 7\times$ too large, so the second-order energy came out ≈ -11 eV ($\sim 30\times$ too deep) and the DiagMC diagram order ran away (no order-0 samples, the estimator's normalization sector). The bug hid because the *oracle* used for the SVD cross-check made the same omission, and the `.epb` test validates only the *Cartesian* coupling, above the projection. *Fix:* apply $(2M_\kappa\omega_{\mathbf{q}\nu})^{-1/2}$ in the branch projection of *both* the tabulated vertex and the direct oracle, guarding the $\omega \to 0$ acoustic modes at Γ. *Validated:* $E^{(2)}$ fell to -0.10 eV on the 8^3 grid — the expected $\sim 3.5\times$ coarse-grid underestimate of the continuum $-\alpha\hbar\omega_{\mathrm{LO}} \approx -0.37$ eV — and the sampler then equilibrates at order ≈ 3.

P9 — Reading E_0 from the decay estimator: μ and $\tau_{\max}$ (Phase B). The ground state is the large-τ slope of $G(\tau) \sim Z\,e^{-(E_0-\mu)\tau}$, and two knobs must bracket it. The chemical potential μ must sit *just below* E_0: if $\mu > E_0$ the weight grows with τ and the diagram order runs away (symptom: "no order-0 samples"); if μ is far below E_0 the phonon cloud is starved and the fit window falls where G is exponentially small, so E_0 scatters with underestimated error bars. The window must also satisfy $\tau_{\mathrm{fit}} \gg 1/(E_1 - E_0) \sim 1/\omega_{\mathrm{LO}}$ to clear excited-state contamination. The acceptance criterion is that the extracted $E_0 = \mu + \text{slope}$ is μ-*independent*. On the coarse 8^3 grid the polaron is effectively strong-coupling (order 3–5, $E_0 \approx -0.26$ eV), so $\mu \approx -0.5$ eV with $\tau_{\max} \approx 30$ eV^{-1} is the working regime.

9 The finite-β thermodynamic energy estimator (FEP-DMC)

The energy in Section 8 (P9) is read from the large-τ decay of $G(\tau)$, which carries a documented sensitivity to the fit window $\tau_{\max}$ at strong coupling. FEP-DMC (Luo, Park, Bernardi, *Nat. Phys.* **21**, 1275 (2025); code a Perturbo fork) instead measures the polaron energy as a *thermodynamic* (operator) expectation value on each worldline, with *no* fit. This section records how that estimator works and its single-band form. It is **implemented** as the production estimator: the sampler runs in a fixed-τ ensemble (external time pinned at $\tau_{\max}$), where the chemical potential μ drops out of the ensemble entirely — eliminating the μ-bracketing of P9 — and the local energy $E_{\mathrm{loc}} = \tau^{-1}\big[\sum_{\mathrm{seg}} \varepsilon\,\Delta\tau + \sum_{\mathrm{arcs}} \omega\,\Delta\tau - 2N_{\mathrm{arcs}}\big]$ is averaged with the sign. It is validated against an exactly solvable dispersionless (Lang–Firsov) lattice model, $E_0 = -\bar{g}^2/\omega$ at all couplings, and against the Fröhlich $\alpha = 1, 2$ benchmarks, where it agrees with the decay fit at lower variance.

Quantity measured. Rather than the asymptotic slope, the polaron energy is the operator average

$$E_{\mathrm{pol}} \;=\; \langle \hat{H} \rangle \;=\; \frac{\mathrm{Tr}\big[\hat{H}\, e^{-\beta\hat{H}}\big]}{\mathrm{Tr}\big[e^{-\beta\hat{H}}\big]}, \tag{5}$$

evaluated on each Monte-Carlo worldline of total imaginary-time length $\tau = \beta$, where $\beta = 1/k_B T$ is the code's `tauMax`. Here Tr is the matrix trace $\mathrm{Tr}[M] = \sum_i M_{ii}$; the trace over the complete basis is what turns Eq. (5) into a proper thermal (and, as $\beta \to \infty$, ground-state) expectation value. This is the standard Prokof'ev–Svistunov / Mishchenko thermodynamic estimator.

Worldline weight. A configuration is one electron backbone on $\tau \in [0, \beta]$, cut into segments by vertices at $\tau_1 < \tau_2 < \cdots$, with phonon arcs joining vertex pairs. In single-band form its weight is a product of

- electron segments s of duration $\Delta\tau_s$: $e^{-\varepsilon_{k_s}\Delta\tau_s}$ (the code's `Gel`$(\varepsilon, \tau) = e^{-\varepsilon\tau}$ and `propagator`). The electron is kept at $T = 0$; only phonons are thermal, as appropriate for a single carrier;
- phonon arcs a of span $\Delta\tau_a$ at frequency ω_a: the finite-T propagator $D(\omega, \tau)$ below;
- a vertex factor $g_{mn\nu}(\mathbf{k}, \mathbf{q})$ at each of the N_v vertices.

Thus $\ln w$ depends on β through every segment length, every arc span, and N_v.

The estimator as a time average. With $\hat{H} = \hat{H}_{\rm el} + \hat{H}_{\rm ph} + \hat{H}_{\rm int}$, the key move is *not* to read the energy only at the endpoint $\tau = \beta$ but to use the ground-state-dominated identity

$$\langle \hat{H} \rangle \;=\; \frac{1}{\beta} \int_0^\beta \langle \hat{H}(\tau) \rangle \, d\tau, \tag{6}$$

i.e. the *local* energy is averaged over the entire worldline. This is the variance reduction: β times more information per configuration than a single asymptotic slope. Each term of $\hat{H}$ maps to one accumulator, and the three sum to

$$E_{\rm pol} \;=\; \frac{1}{\beta \, \langle s \rangle} \Big\langle \underbrace{\sum_{\rm seg} (\varepsilon_i + \mu)\, \Delta\tau_i}_{\langle H_{\rm el} \rangle} \;+\; \underbrace{\sum_{\rm arcs} \tfrac{1}{2}\, \omega_a |\Delta\tau_a|}_{\langle H_{\rm ph} \rangle} \;-\; \underbrace{(N_v - 1)}_{\langle H_{\rm int} \rangle} \Big\rangle, \tag{7}$$

with $\langle s \rangle$ the average sign. The three pieces are:

Band energy $\langle H_{\rm el} \rangle$. The code inserts the operator via `Hpropagator`, whose diagonal is $e^{-\varepsilon\tau}(\varepsilon + \mu)\tau$. The extra factor $\varepsilon\,\Delta\tau$ is exactly $\int_0^{\Delta\tau} ds\, \varepsilon\, e^{-\varepsilon\Delta\tau} = \varepsilon\,\Delta\tau\, e^{-\varepsilon\Delta\tau}$ — the "$\hat{H} e^{-\hat{H}\tau}$" insertion integrated over the segment. Summed over segments and divided by the diagram weight it yields $\frac{1}{\beta}\int_0^\beta \varepsilon_{k(\tau)}\, d\tau$. Energies are stored band-bottom subtracted (ε) with $\mu = \min \varepsilon$, so $\varepsilon + \mu$ restores the true band energy.

Phonon energy $\langle H_{\rm ph} \rangle$. Each arc present over $\Delta\tau_a$ contributes ω_a to the instantaneous phonon energy; time-averaging gives $\frac{1}{\beta}\sum_a \omega_a \Delta\tau_a$. The $\frac{1}{2}$ corrects double counting of an internal arc seen from both its vertices; external (head/tail-anchored) phonon lines carry no $\frac{1}{2}$.

Interaction energy $\langle H_{\rm int} \rangle$. In the bare imaginary-time expansion the coupling-energy estimator evaluates to $-\langle N_v \rangle/\beta$ (each vertex contributes $-1/\beta$); the "-1" offset is the head-vertex normalization convention.

Equation (7) is thus just $\langle H_{\rm el} \rangle + \langle H_{\rm ph} \rangle + \langle H_{\rm int} \rangle = \langle \hat{H} \rangle$, measured at $\tau = \beta$ with no fitting window.

Finite-β phonon propagator. The arcs use the thermal bosonic propagator

$$D(\omega, \tau) \;=\; \frac{e^{-\omega\tau} + e^{-\omega(\beta-\tau)}}{1 - e^{-\omega\beta}}, \tag{8}$$

whose first term $e^{-\omega\tau}$ is spontaneous emission (surviving at $T = 0$), second term $e^{-\omega(\beta-\tau)}$ is thermal absorption / wrap-around, and denominator $1/(1 - e^{-\omega\beta}) = 1 + n_B(\omega)$ is the Bose enhancement.

As $\beta \to \infty$ it collapses to $e^{-\omega\tau}$, which is what the code's `zeroTMC` flag hard-codes — and the LiF example uses `zeroTMC`, so LiF(e) is effectively the $T = 0$ limit even though the machinery is finite-T. Soft modes below a `phfreq_cutoff` ≈ 1 meV are dropped ($D \to 0$) to avoid the $1/(1 - e^{-\omega\beta})$ and $(2M_\kappa\omega)^{-1/2}$ divergences.

Matrix (multiband) form. In multiband mode the segment propagators, vertices, and the $\hat{H}$-insertion are all matrices, so the diagram weight is $\mathrm{Tr}\big[\prod_i g_i\, e^{-H\Delta\tau_i}\big]$ and inserting $\hat{H}$ at one segment needs the product of everything to its left and right (the `left_env`/`right_env` environment matrices), with per-step rescaling. The sign is carried as $\mathrm{sign}(\mathrm{Re}\,\mathrm{Tr})$ and divided out via $\langle s\rangle$. In single band all of this collapses to the scalars of Eq. (7); the trace is trivial.

Why it replaced the decay fit. Relative to the decay fit of P9, Eq. (7) removes the $\tau_{\max}$ window arbitrariness (the whole $[0, \beta]$ worldline is used), lowers variance (it time-averages rather than reading one slope), and has a systematic error controlled by β (excited-state contamination $\sim e^{-\Delta\beta}$). The single-band form accumulates the three sums of Eq. (7) per sample, dividing by β and $\langle s\rangle$ — no environment matrices are needed until the multiband (hole/degenerate) work of D2. It has been cross-validated against the Fröhlich $\alpha = 1, 2$ Mishchenko values (rung 0), where it agrees with the decay estimator, and against the exactly solvable dispersionless lattice model noted above; all production numbers (Section 6) use it.